AI-Based Predictive Analytics for Diabetes Complications

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1 Introduction

Diabetes mellitus is a chronic metabolic disorder associated with severe long-term complications, including kidney disease, neuropathy, and cardiovascular conditions. Early identification of high-risk patients can improve disease management and reduce healthcare costs.

This project develops a machine learning-based predictive analytics system to identify patients at risk of developing multiple diabetes-related complications from clinical data.

2 Problem Definition

Traditional rule-based systems rely on static thresholds and isolated clinical measurements. In practice, diabetes complications develop gradually and are influenced by interactions among medication usage, laboratory results, and hospitalization patterns.

Machine learning models can capture these complex relationships and generate predictive insights that support earlier clinical intervention.

3 Dataset Description

This project uses the *Diabetes 130-US Hospitals* dataset, which contains over 100,000 hospital encounters for diabetic patients from 130 hospitals (1999–2008).

The dataset includes the following information:

- Patient demographics
- Laboratory test results
- Medication usage
- Hospital visit information

These variables are used to predict the likelihood of three diabetes-related complications:

- Cardiovascular complications
- Kidney disease
- Diabetic neuropathy

4 Data Acquisition

The dataset was obtained from the UCI Machine Learning Repository and downloaded locally for preprocessing and analysis. Python scripts were used to load, validate, and prepare the

dataset for downstream modeling.

5 Data Cleaning

Several preprocessing steps were performed to prepare the dataset for machine learning:

- Missing values represented by "?" were converted to NaN.
- Columns with excessive missingness (for example, weight, payer code, and medical specialty) were removed.
- Diagnosis codes were cleaned and standardized.
- Laboratory test variables were encoded to represent test outcomes.

6 Feature Engineering

Additional features were generated to support multi-label classification of diabetes complications.

Binary target labels were created to indicate the presence of:

- Cardiovascular complications
- Kidney disease
- Neuropathy

These labels allow the predictive system to identify multiple complications within a single patient encounter.

7 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand dataset structure and identify clinically meaningful patterns.

7.1 Complication Distribution

Figure 1 presents the distribution of the three complication categories.

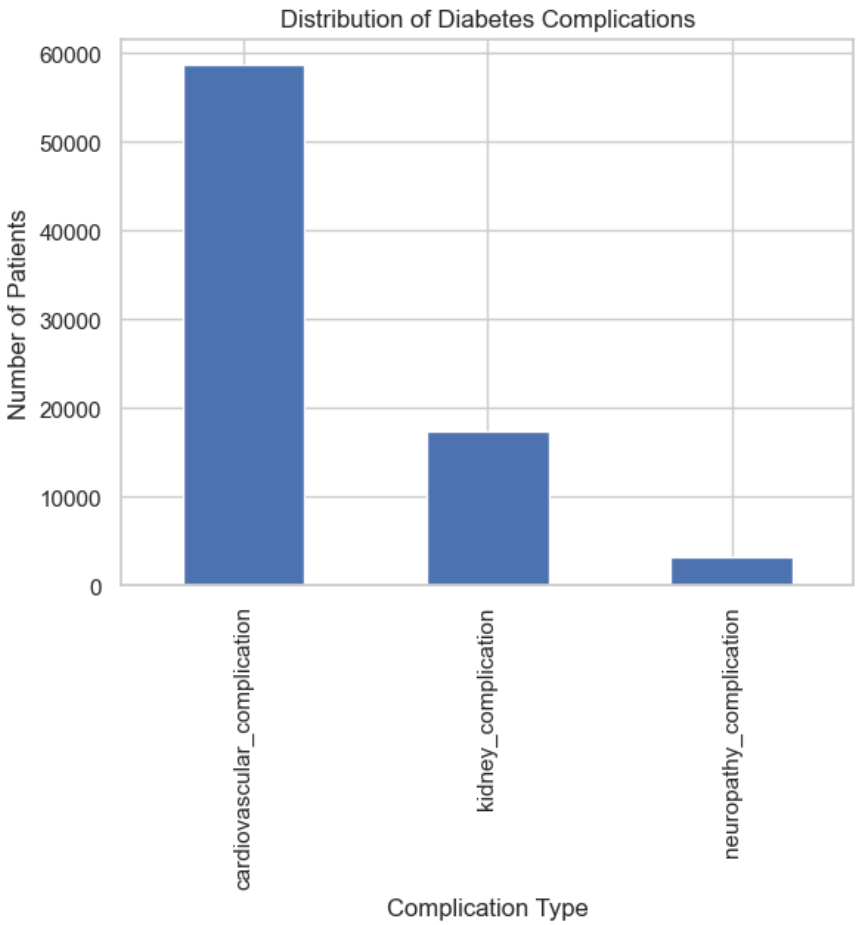

Figure 1: Distribution of Diabetes Complications

7.2 Age Distribution

Figure 2 illustrates the patient age distribution in the dataset.

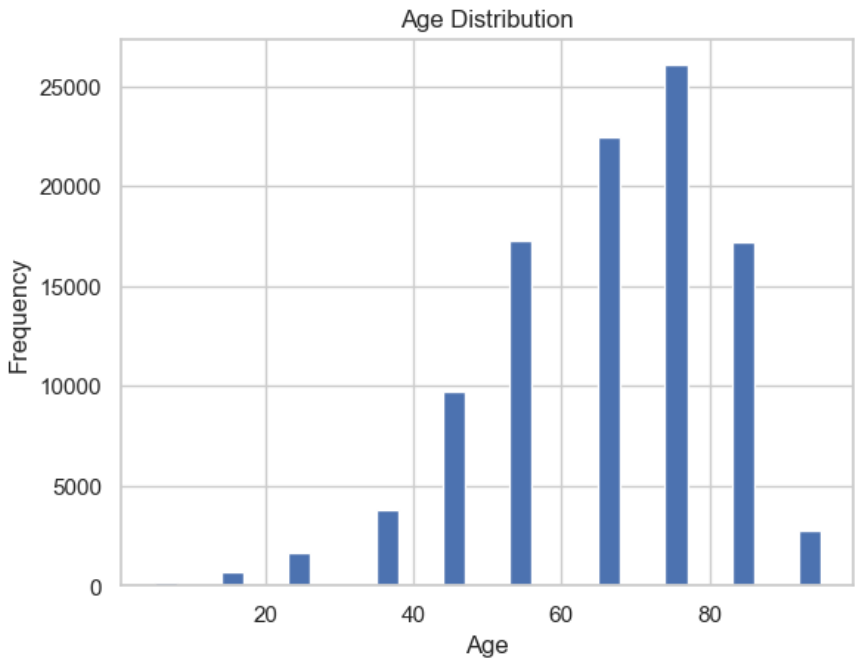

Figure 2: Age Distribution of Patients

7.3 Feature Correlation

Figure 3 presents a correlation heatmap of relationships among clinical features.

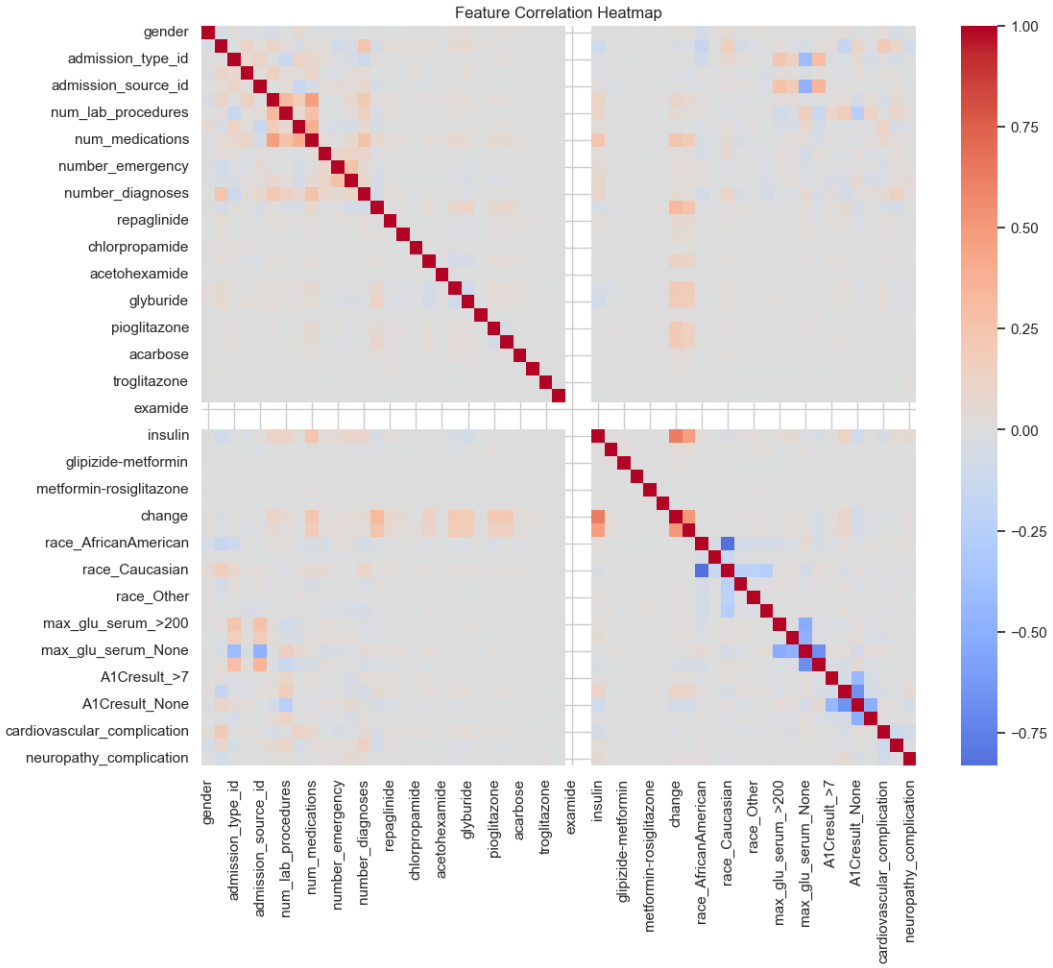

Figure 3: Correlation Between Dataset Features

7.4 Complication Correlation

Figure 4 shows how different complications co-occur.

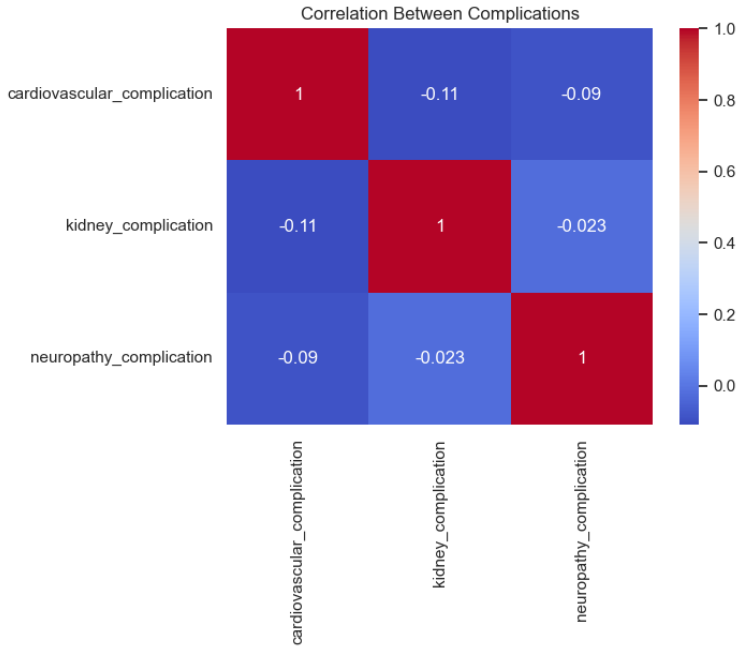

Figure 4: Correlation Between Diabetes Complications

7.5 Hospital Visit Patterns

Figure 5 summarizes the average number of outpatient, emergency, and inpatient visits.

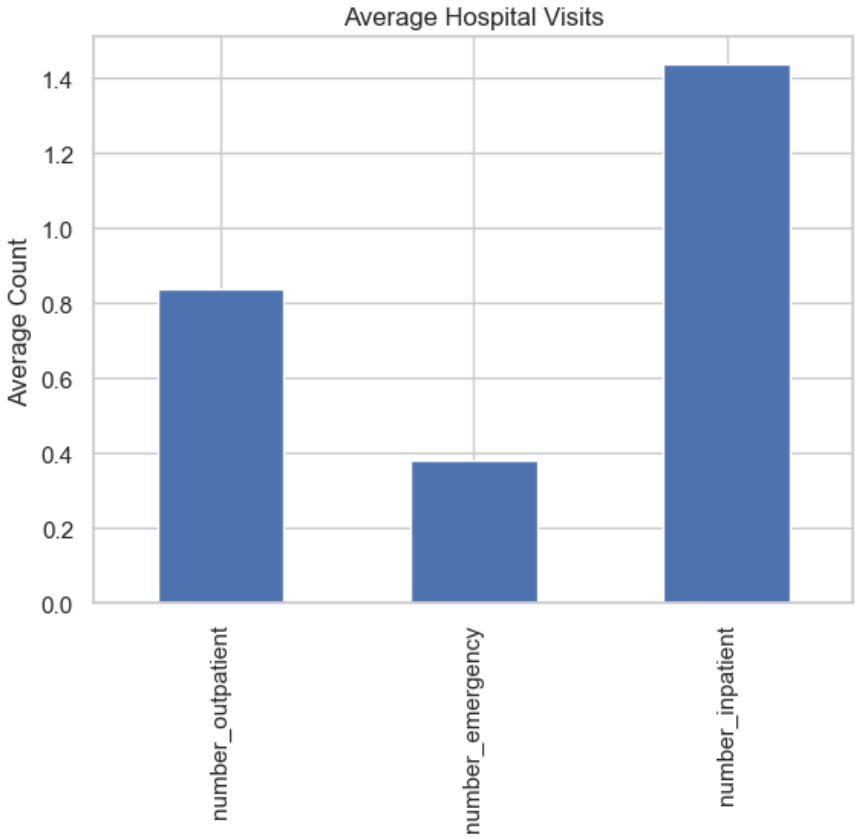

Figure 5: Average Hospital Visits

7.6 Insulin Usage

Figure 6 shows the distribution of insulin treatment levels across patients.

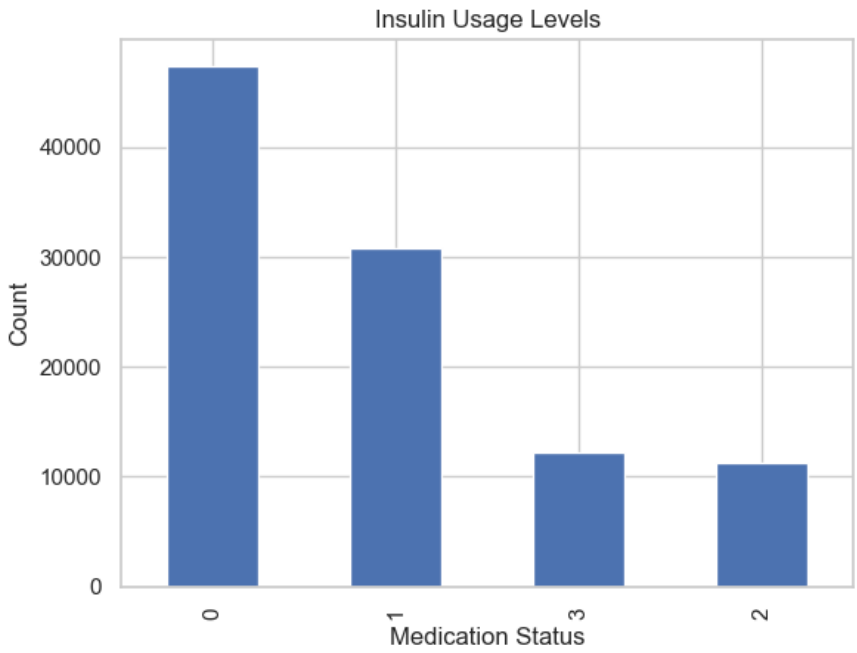

Figure 6: Distribution of Insulin Usage

8 Conclusion

This deliverable defined the problem scope and prepared the dataset for machine learning. Data preprocessing and exploratory analysis transformed raw clinical records into a structured dataset suitable for predictive analytics.

Future work will focus on training multi-label machine learning models and evaluating performance using sensitivity and F1-score metrics.